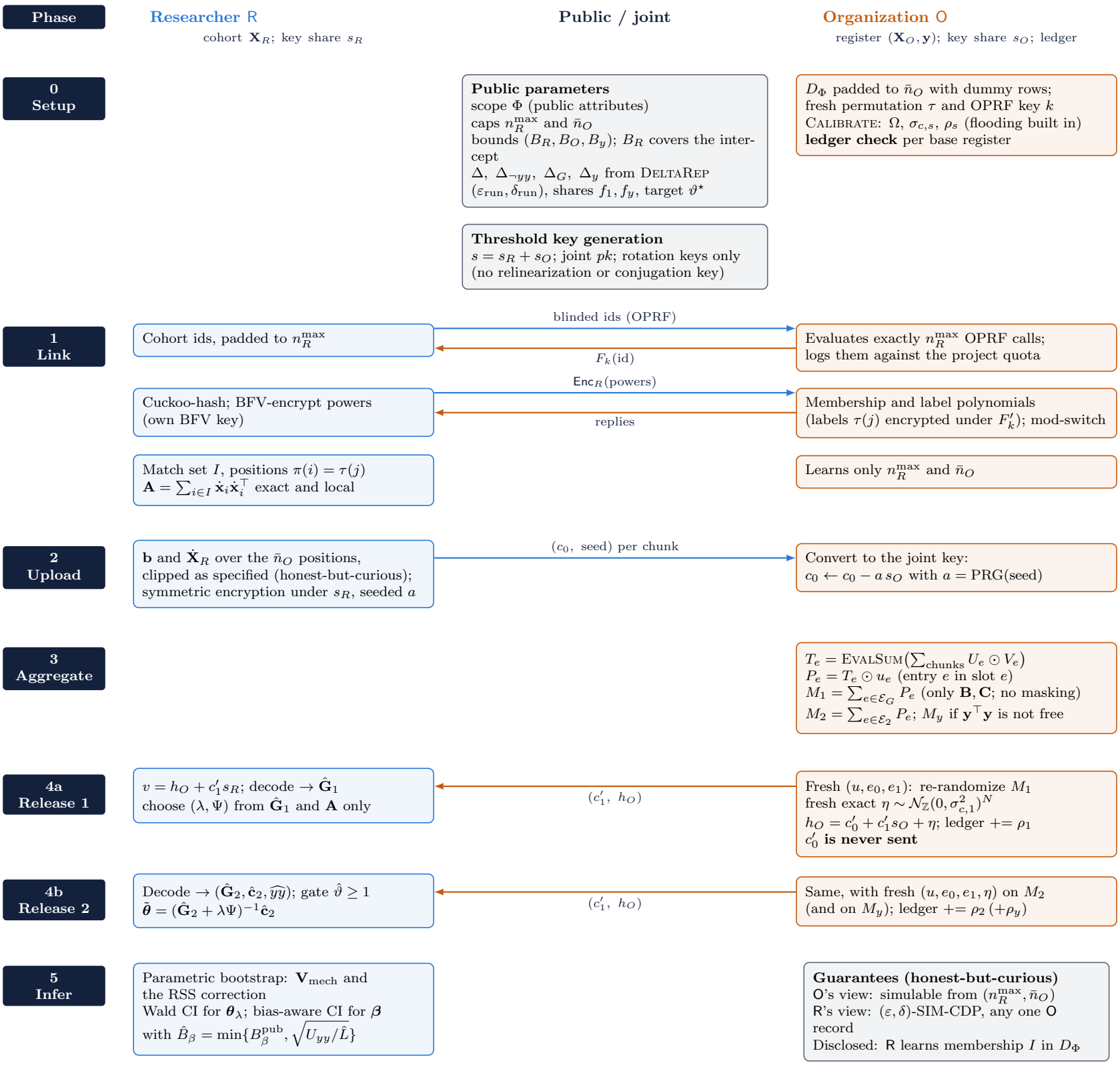

Phase
Researcher R
cohort $\mathbf{X}_R$; key share s_R
Public / joint
Organization O
register $(\mathbf{X}_O, \mathbf{y})$; key share s_O; ledger
0 Setup
Public parameters
scope Φ (public attributes)
caps $n_R^{\max}$ and $\bar{n}_O$
bounds (B_R, B_O, B_y); B_R covers the intercept
Δ, $\Delta_{\neg yy}$, Δ_G, Δ_y from DeltaRep
$(\varepsilon_{\text{run}}, \delta_{\text{run}})$, shares f_1, f_y, target $\vartheta^\star$
D_Φ padded to $\bar{n}_O$ with dummy rows;
fresh permutation τ and OPRF key k
Calibrate: Ω, $\sigma_{c,s}$, ρ_s (flooding built in)
ledger check per base register
Threshold key generation
$s = s_R + s_O$; joint pk; rotation keys only
(no relinearization or conjugation key)
1 Link
Cohort ids, padded to $n_R^{\max}$
blinded ids (OPRF)
$F_k(\text{id})$
Evaluates exactly $n_R^{\max}$ OPRF calls;
logs them against the project quota
Cuckoo-hash; BFV-encrypt powers
(own BFV key)
$\text{Enc}_R(\text{powers})$
replies
Membership and label polynomials
(labels $\tau(j)$ encrypted under F'_k); mod-switch
Match set I, positions $\pi(i) = \tau(j)$
$\mathbf{A} = \sum_{i \in I} \dot{\mathbf{x}}_i \dot{\mathbf{x}}_i^\top$ exact and local
Learns only $n_R^{\max}$ and $\bar{n}_O$
2 Upload
$\mathbf{b}$ and $\dot{\mathbf{X}}_R$ over the $\bar{n}_O$ positions,
clipped as specified (honest-but-curious);
symmetric encryption under s_R, seeded a
$(c_0,$ seed) per chunk
Convert to the joint key:
$c_0 \leftarrow c_0 - a\, s_O$ with $a = \text{PRG}(\text{seed})$
3 Aggregate
$T_e = \text{EvalSum}(\sum_{\text{chunks}} U_e \odot V_e)$
$P_e = T_e \odot u_e$ (entry e in slot e)
$M_1 = \sum_{e \in \mathcal{E}_G} P_e$ (only $\mathbf{B}, \mathbf{C}$; no masking)
$M_2 = \sum_{e \in \mathcal{E}_2} P_e$; M_y if $\mathbf{y}^\top \mathbf{y}$ is not free
4a Release 1
$v = h_O + c'_1 s_R$; decode $\to \hat{\mathbf{G}}_1$
choose (λ, Ψ) from $\hat{\mathbf{G}}_1$ and $\mathbf{A}$ only
$(c'_1,\ h_O)$
Fresh (u, e_0, e_1): re-randomize M_1
fresh exact $\eta \sim \mathcal{N}_{\mathbb{Z}}(0, \sigma_{c,1}^2)^N$
$h_O = c'_0 + c'_1 s_O + \eta$; ledger += ρ_1
c'_0 is never sent
4b Release 2
Decode $\to (\hat{\mathbf{G}}_2, \hat{\mathbf{c}}_2, \widehat{yy})$; gate $\hat{\vartheta} \geq 1$
$\tilde{\boldsymbol{\theta}} = (\hat{\mathbf{G}}_2 + \lambda\Psi)^{-1} \hat{\mathbf{c}}_2$
$(c'_1,\ h_O)$
Same, with fresh (u, e_0, e_1, η) on M_2
(and on M_y); ledger += ρ_2 (+ρ_y)
5 Infer
Parametric bootstrap: $\mathbf{V}_{\text{mech}}$ and
the RSS correction
Wald CI for $\boldsymbol{\theta}_\lambda$; bias-aware CI for $\boldsymbol{\beta}$
with $\hat{B}_\beta = \min\{B_\beta^{\text{pub}}, \sqrt{U_{yy}/\hat{L}}\}$
Guarantees (honest-but-curious)
O's view: simulable from $(n_R^{\max}, \bar{n}_O)$
R's view: (ε, δ)-SIM-CDP, any one O record
Disclosed: R learns membership I in D_Φ